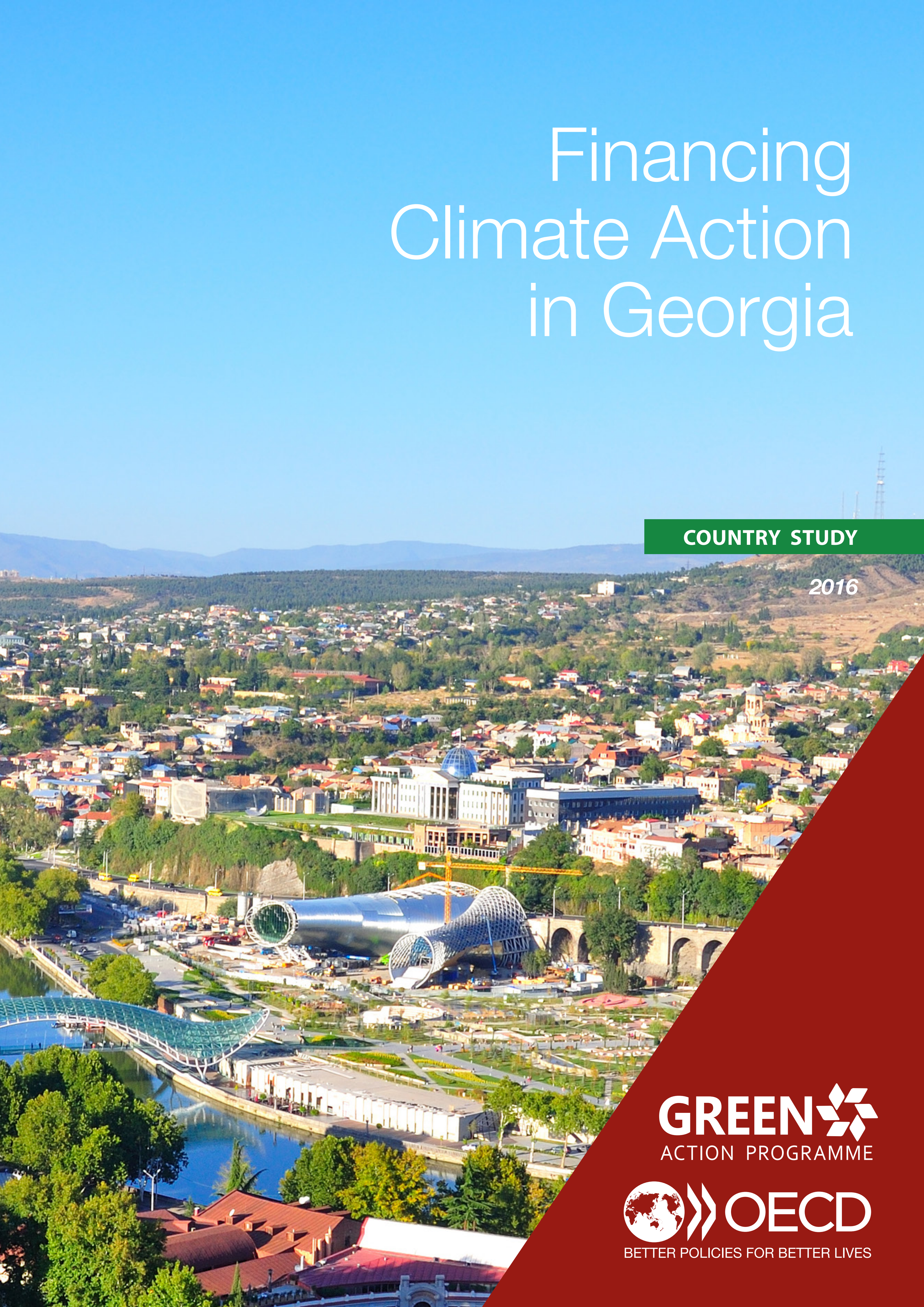


Financing Climate Action in Georgia

COUNTRY STUDY

2016

GREEN ACTION PROGRAMME

OECD
BETTER POLICIES FOR BETTER LIVES

Summary

Georgia submitted its intended nationally determined contributions (INDC) in 2015, highlighting the need for addressing both climate change adaptation and mitigation. Through the INDC, Georgia communicated its intention to reduce greenhouse gas (GHG) emissions by at least 15% below the business-as-usual scenario by 2030. Georgia has been developing a range of legal and policy frameworks, relating to climate change and the wider sustainable development agenda (e.g. the Law on Electricity and the Natural Gas and National Energy Efficiency Action Plan). Georgia is also developing its Low Emission Development Strategy (LEDS) and the National Energy Efficiency Action Plan (NEEAP). Multiple nationally appropriate mitigation actions (NAMAs) are also being developed on: energy efficient refurbishment in the public building sector; efficient use of biomass for equitable, climate proof and sustainable rural development; and urban mobility (Mdivani, 2016).

During the period 2013-14, approximately USD 239 million per year of climate-related development finance was committed to support mitigation and adaptation actions in Georgia, while the amounts fluctuated considerably between these two years. The level of the committed amount was lower than the average among the countries of Eastern Europe, the Caucasus and Central Asia (EECCA) (i.e. USD 303 million per year). However, annual climate-related development finance "per capita" committed to the country (approximately USD 55 per capita per year) was about twice larger than the EECCA average (USD 27 per capita annually).

The largest amount of climate-related development finance in 2013 and 2014 was committed to the energy sector (i.e. 67.8% of total). Examples of large-scale energy projects include the development or rehabilitation of hydropower plants and energy efficiency in transmission networks. There have been projects on other types of renewable energy (e.g. biomass and wind energies), and energy efficiency on demand side over the past five years. With regard to adaptation, climate-related development finance was committed mostly to the forestry and agriculture sectors and disaster risk management.

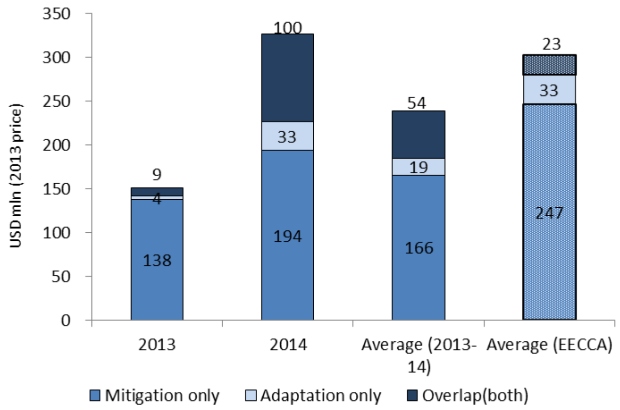
Most of the climate-related finance committed in 2013 and 2014 is delivered through multilateral development banks (67%: e.g. the European Bank for Reconstruction and Development and the International Finance Corporation), followed by bilateral donors (32%: e.g. Germany, the EU and France). Loans are predominantly used as financial instruments.

Within the country, the Ministry of Environment and Natural Resource Protection is the national focal point, or designated authority, to the UNFCCC, the Green Climate Fund and the Global Environment Facility, and is involved in a range of climate-related projects that are supported by international sources. However, many other ministries and governmental agencies such as the Ministry of Economy and Sustainable Development also engage in such projects.

Overview of climate-related development finance to Georgia in 2013-2014: Excerpt from the report

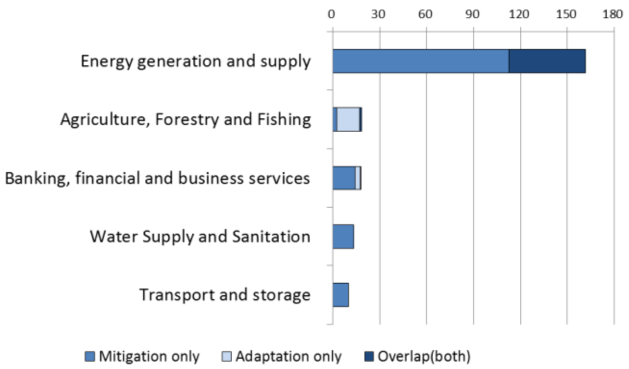
Total climate-related development finance flows by activities (mitigation, adaptation, and both)

(USD million per year)



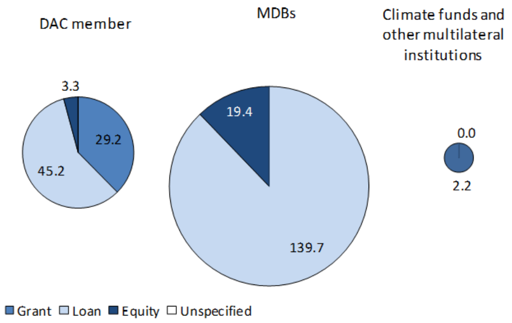
Climate-related development finance flows by sector

(USD million per year)



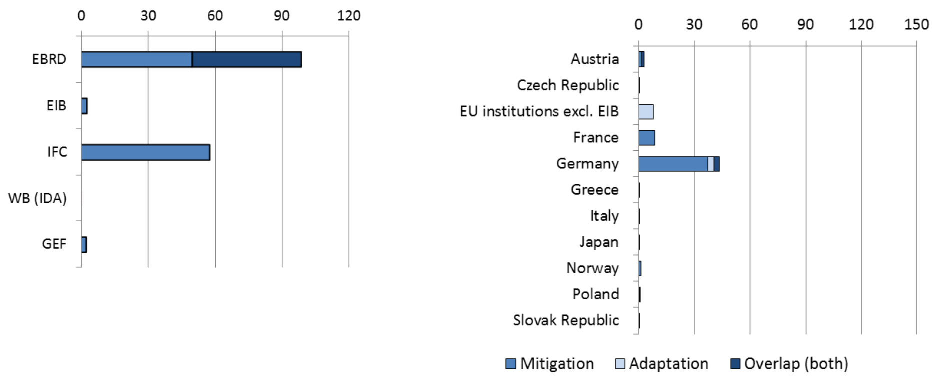
Financial instruments used by delivery channel

(USD million per year)



Major providers of climate-related development finance

(USD million per year)



Note 1: Total climate-related development finance = Mitigation + Adaptation – Overlap (both).
Note 2: Names of the sectors correspond to those used in the DAC CRS database.
Note 3: Please see the 'Reader's guide' section for more information on methodological approach

Source: Based on OECD (2016)

This country-level study complements *OECD (2016),"Financing Climate Action in Eastern Europe, the Caucasus and Central Asia*", and was prepared as part of the project "International Climate Finance for EECCA" under the GREEN Action Programme hosted by the Organisation for Economic Co-operation and Development (OECD). The project has been implemented with support of the German Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety. The report benefitted from the discussions at the Expert Workshop on International Climate Finance for EECCA that was held on 11 July 2016 in Paris, and written comments provided by the participants before and after the workshop.

The views expressed herein can in no way be taken to reflect the official opinion of Germany, or any of the OECD member countries, or the endorsement of any approach described herein. This document is also without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

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Information on the OECD-hosted GREEN Action Programme and relevant publications is available at http://www.oecd.org/env/outreach/eap-tf.htm

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Reader's guide

This country-level study aims to provide an overview of how international development finance has been supporting climate-related actions in the recent years, so as to improve clarity on how each of the countries in Eastern Europe, the Caucasus and Central Asia (EECCA) region is working together with their development co-operation partners. The study is based on both:

(i) quantitative analysis for the period between 2013 and 2014; and

(ii) qualitative analysis during the period between 2011 and 2015.

The 11 country reports were prepared to complement a publication "Financing Climate Action in Eastern Europe, Caucasus and Central Asia" by the Organisation for Economic Co-operation and Development (OECD) (Available at http://www.oecd.org/env/outreach/eap-tf.htm).

This study does not offer a complete picture of climate finance from all possible sources in public and private sectors, or all relevant policy frameworks within the country. However, it intends to provide a clearer understanding of international (public) financing flows committed to each of the 11 EECCA countries in terms of major sectors/areas, providers, and financing structures for individual projects, as well as domestic institutions involved in accessing and using such finance, on which relevant data tend to be scattered.

The study also analyses the country's climate targets and priority sectors/areas for climate actions based on its INDC and/or other relevant policy documents. Finally, the study briefly outlines in-country enabling environments, such as policies, laws, institutional arrangements and domestic financing mechanisms, which aim to promote a low-carbon, climate-resilient development.

The quantitative analysis for the period 2013-14 is conducted by using the database from the OECD DAC Creditor Reporting System (CRS)[1]. This database allows for an approximate quantification of climate-related development finance flows that target climate mitigation and adaptation as either their principle objective or significant objective. The bilateral sources include OECD DAC members, while multilateral sources include multilateral development banks and international climate funds. Some of the South-South co-operation and non-DAC member contributions are also included.

The qualitative analysis for the period 2011-15 is based on publicly available project-level information (e.g. project design documents, project evaluation reports, and periodic reports by donors and financial institutions). In this part, sizes of some projects are indicated as committed financing volumes for the entire projects, while for reporting purpose, multilateral development banks only report the value of the components specifically relating to climate action as climate finance.

The DAC CRS records face values of the activities on the dates when grant or loan agreements are signed with recipients (i.e. commitment, but not disbursement). It should also be noted that the scope of the data sources for both the quantitative and qualitative analyses do not include some of the non-DAC member donors such as the People's Republic of China and the Russian Federation, or private sector investors, whose financial provision may be significant in certain EECCA countries.

The cut-off date of inclusion of information on data, policies and projects was 01 August 2016.

[1] For more details, see http://www.oecd.org/dac/stats/climate-change.htm and on the DAC members see http://www.oecd.org/dac/dacmembers.htm.

Background

Georgia is a Lower Middle Income Country, with USD 7 233 per capita GDP purchasing power parity (PPP) and a population of 3.6 million in 2014 (WB, 2016b). The country achieved a steady economic growth between 2003 and 2014 with a 6% annual growth rate, due largely to structural policy and market reforms that stimulated capital inflows and investment, while the growth slowed to 2.8% in 2015 due to weak external environments (WB, 2016). These reforms helped Georgia improve its business environment, strengthened public finance systems, upgraded infrastructure and liberalised trade.

The World Bank projected that Georgia's annual economic growth will be 5.5% on average over the medium term, dependent on greater policy certainty, improved market access, and on strong structural reform implementation and attaining benefits from the Deep and Comprehensive Free Trade Area and Association Agreement with the European Union (WB, 2016a).

Georgia emitted approximately 16 million tCO_2e of greenhouse gases (GHG) in 2012 (GoG, 2015a), which is about 0.03% of the global GHG emissions. Georgia is a non-Annex I country to the United Nations Framework Convention on Climate Change (UNFCCC). Approximately, 82% of the GHG emissions was from energy use, followed by agriculture (28%). 30% of energy-related GHG emissions comes from the transport sector, and 24% is from "other fuel combustion" (mainly in residential and commercial buildings). Electricity and heat generation accounts for 16% while the industry and manufacturing sector accounts for 17% of the energy-related GHG emissions (WRI, 2016).

Adverse impacts of climate change are considered to be severe threats to Georgia's sustainable development (GoG, 2015a). In particular, sea level rise has affected certain areas of land near the Black Sea, and destroyed or damaged buildings and infrastructure along the coast, which can affect negatively the country's tourism industry. Increased frequency and intensity of floods, landslides and mudflows have caused considerable damages in the economy of highland areas. Further, rising temperatures, increased winds and reduced water availability have caused a significant decline in agricultural productivity, while decreased rainfall and enhanced evaporation are imposing risks of desertification in the semi-arid regions in the eastern part of Georgia (GoG, 2015b).

Targets and priority areas for climate actions

In its intended nationally determined contribution (INDC), Georgia has communicated conditional (on international support) and unconditional GHG emission reduction targets. Its unconditional target is to reduce GHG emissions by 15% below the business as usual (BAU) scenario towards 2030, while the conditional target that assumes international financial support and technology transfer is to reduce GHG emissions by 25% by 2030 (GoG, 2015a). It is calculated that, in absolute terms, a 15% reduction below BAU would mean about 32% below the 1990 level, and a 25% below BAU would mean approximately 41% below the 1990 level.

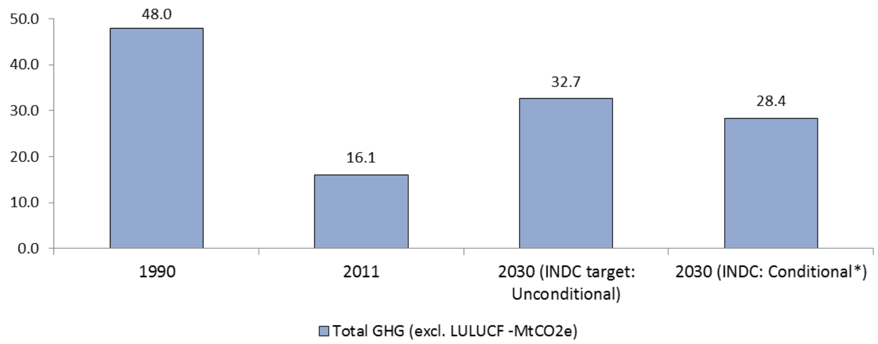
While the INDC does not include information on specific actions to be taken to achieve the goals, it refers to other official policy documents on climate mitigation actions, namely the Low Emission Development Strategy (LEDS) and the National Energy Efficiency Action Plan (NEEAP). Georgia has also been developing multiple nationally appropriate mitigation actions (NAMAs) for: energy efficient refurbishment in the public building sector; efficient use of biomass for equitable, climate proof and sustainable rural development; and urban mobility (Mdivani, 2016). These NAMAs are also planned to be linked with the NEEAP (ibid.).

Georgia plans to finalise LEDS in 2016 with the support of the United States Agency for International Development (USAID). The LEDS aims to detail the pre-2020 mitigation actions. Further, the government is also planning to finalise the NEEAP by the early 2016[2], which documents the plans for implementation of energy efficiency measures (Arabidze, 2016). The first draft for NEEAP outlines a range of energy efficiency measures, such as the reduction of losses in energy generation, transmission and consumption, reduction of losses in gas pipelines, efficient lighting and designing for industrial facilities and residential buildings, saving energy through industrial processes, energy efficient transport sector. These energy efficiency measures as well as the stimulation of a range of renewable energy sources (e.g. wind, biomass, solar and geothermal) are also identified in the Technology Needs Assessment that was finalised in 2012 (GoG, 2012).

It is also notable that 15 self-governing cities and municipalities in Georgia have signed the European Union's "Covenant of Mayors" initiative and plan to voluntarily reduce their own GHG emissions. This initiative is expected to significantly contribute to the post-2020 implementation processes of Georgian climate mitigation actions (GoG, 2015a). As of the end-2015, 13 cities are signatories of the Covenant of Mayors, of which eight cities have submitted their Sustainable Action Plans (Abulashvili, 2016)

The adaptation section of the INDC mentions agriculture, disaster risk management and coastal protection against the sea level rise of the Black Sea. The INDC also outlines types of technologies for which international support is needed, such as technologies for: the protection of coastal infrastructure; sustainable water management; sustainable agriculture; and sustainable forest management. The implementation of adaptation actions will be further elaborated in the country's national adaptation plan (NAP) that is currently being developed. In addition, the country's latest National Communication to the UNFCCC also identifies agriculture, healthcare and tourism as important sectors to be evaluated for their vulnerability to climate change (GoG, 2015b).

Figure 1. GHG emissions in the base year, the recent data and the target year



Sources: GoG (2015) Georgia's Intended National Determined Contribution.

[2] As of March 2016, the final version of NEEAP is planned to be launched in May-June 2016 (GoG, 2016).

Table 1. Summary of the INDC

Scope of action	Targets	Priority sectors for mitigation actions
Mitigation	[Unconditional] To reduce GHG emissions by 15% below the BAU* by 2030	• Energy efficiency and renewable energy on both the supply and demand sides
	[Conditional on international support] To reduce GHG emissions by 25% by 2030	
Adaptation	The main objective is to improve the country's preparedness and adaptive capacity by developing climate resilient practices that reduce vulnerability of highly exposed communities.	• Agriculture • Disaster risk management • Coastal zone protection
Means of implementation	**Quantified needs if any**	**Description**
Finance	(The INDC mentions that the more ambitious target is subject to "technical cooperation, access to low-cost financial resources and technology transfer".)	The INDC does not indicate specific figures of need to be supported by international sources, but indicates that the total adaptation costs would be USD 1.5-2 bln in the period 2021-2030
Capacity development		N.A.
Technology Transfer		Priorities in the needs for international support for technologies transfer are: the protection of coastal infrastructure; technologies for sustainable water management; sustainable agricultural technologies; and technologies for sustainable forest management.

Note: * Business as usual.
Sources: GoG (2015) Intended Nationally Determined Contribution (INDC).

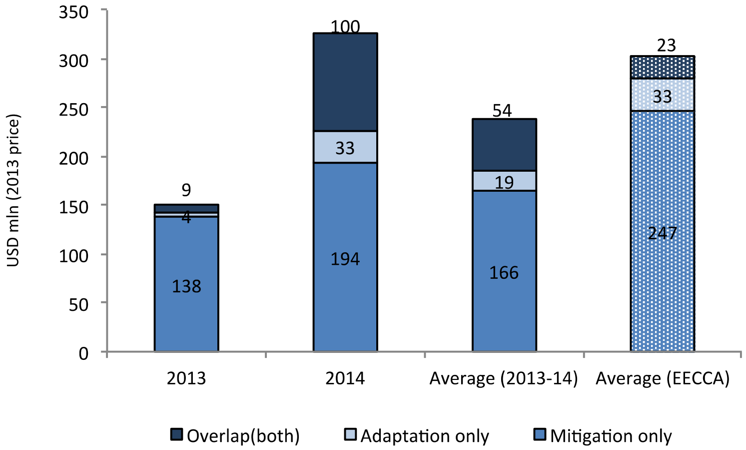
Overview of climate-related development finance flows (2013-14)

USD 238.9 million per year of climate-related development finance was committed to mitigation and adaptation actions in Georgia in the period 2013-14, while the amounts in the two years fluctuated considerably (Figure 2). The hike in the amount in 2014 is partly due to a large-scale project for the Shuakhevi hydropower station construction, supported by the European Bank for Reconstruction and Development (EBRD), which amounted to USD 97.5 million in 2014. This project targets both mitigation and adaptation. Nevertheless, both the number of projects and the average project size committed was larger in 2014 than in 2013, even if the number in 2014 excludes the project in Shuakhevi .

Apart from the Shuakhevi hydropower plant, a significantly larger amount of climate-related development finance is committed to mitigation projects (70% of the total climate-related development finance to the country, or 88% of the total excluding the Shuakhevi project) than adaptation projects in the period 2013-14. Nearly half of the development finance for mitigation (45%) is related to hydropower plants, while other types of large-scale projects include energy transmission, waste management and transport. This may reflect the considerable potential of hydropower in Georgia, of which only 12% has been developed so far, and it can be further explored. The importance of improving efficiency in energy transmission and consumption infrastructure and diversifying fossil fuel sources is another area which needs further attention (IEA, 2015).

The total amount of finance committed to the country in 2013 and 2014 is smaller than the average for all EECCA countries. However, annual climate-related development finance "per capita" committed to the country (approximately USD 55 per capita) is about twice larger than the EECCA average (USD 27 per capita annually). This is partly explained by the fact that Georgia has a relatively small population and the financing to a number of large-scale projects in the energy sector was actually committed in 2013 and 2014.

Figure 2. Climate-related development finance flows in 2013 and 2014 and the 2-year average (Georgia and the EECCA average: USD million per year)



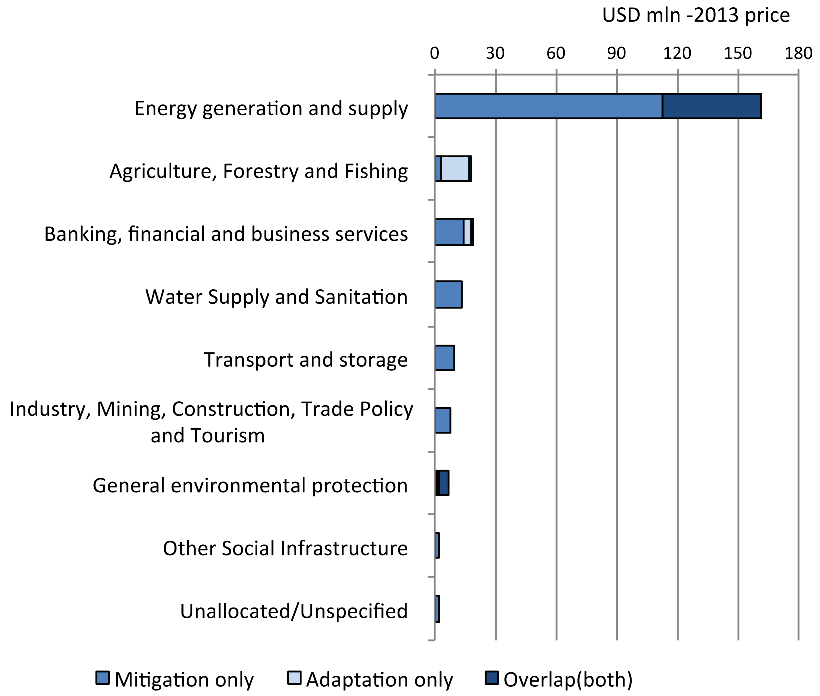
Note: Total climate-related development finance = Mitigation + Adaptation – Overlap (both).
Source: Based on OECD (2016), Climate-related development finance, 2016 February version, Paris.

The energy sector received the largest amount of climate-related development finance in 2013 and 2014 (approximately USD 162 million per year) in Georgia (Figure 3). Examples of large-scale projects include the development or rehabilitation of hydropower plants and energy efficiency in transmission network. As mentioned above, the large share of financing for multi-focal (i.e. both mitigation and adaptation purposes) in the total amount is attributed to the USD 100-billion hydropower project in Shuakhevi. It should be noted that while this project is also financed by the International Finance Corporation (IFC), the Asian Development Bank (ADB) and the government of Canada as well as private sector investors, the IFC reported this project to the OECD DAC Creditor Reporting System as a "mitigation" project. The ADB has not even reported this project as a climate-related project in 2013 or 2014. These examples illustrate the technical challenges and the room for improvement in the tracking exercise on climate-related development finance from various sources.

The second largest amounts of climate-related development finance were committed to the agriculture and forestry sectors as well as the banking sector during the period 2013-14. Climate-related development finance has supported a range of projects in these sectors especially in adaptation. The largest project was the "European Neighbourhood Programme for Agriculture and Rural Development" (USD 15.8 million) that aims to reinvigorate the agricultural sector in Georgia, supported by the European Union (EU). Switzerland also committed to supporting projects that aim to increase the resilience of livestock farming in the rural regions (USD 10.4 million in total).

As for the banking sector, major commitments were made by the EBRD to provide credit lines to local banks for sustainable energy finance (e.g. the Caucasus Energy Efficiency Programme). Germany also provides agricultural credits for micro, small and medium-sized enterprises in the agriculture sector. The United States also supports projects that help improve Georgia's economic governance, particularly in the areas of improving the enabling environments for business development, water resource management and energy trading policy.

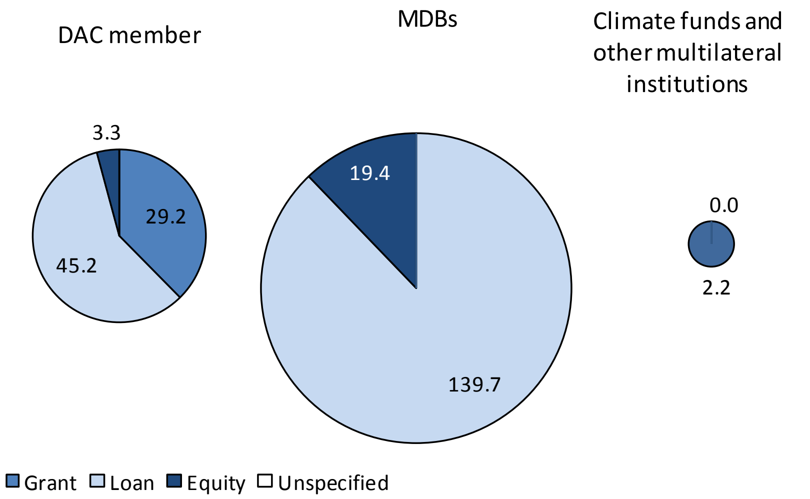
Figure 3. Major sectors to which climate-related development finance is committed in 2013 and 2014 (USD million per year 2013 price)



Note 1: Total climate-related development finance = Mitigation + Adaptation – Overlap (both).
Note 2: Names of the sectors correspond to those used in the DAC CRS database.
Source: Based on OECD (2016), Climate-related development finance, 2016 February version, Paris.

In the period 2013-14, most of the climate-related development finance was committed by multilateral providers such as the EBRD and the IFC, for which loans are predominantly used (Figure 4). Equity financing also supports climate actions in the country. For instance, the IFC provides non-concessional equity financing for a large-scale hydro power development project. Bilateral providers are another major channel and use grants, loans and equities to deliver climate-related development finance. The Global Environment Facility (GEF) is the only dedicated climate fund, recorded in the OECD DAC CRS, which provided funding to Georgia in the period between 2013 and 2014, and uses grant financing to support a range of projects such as on forestry and transport.

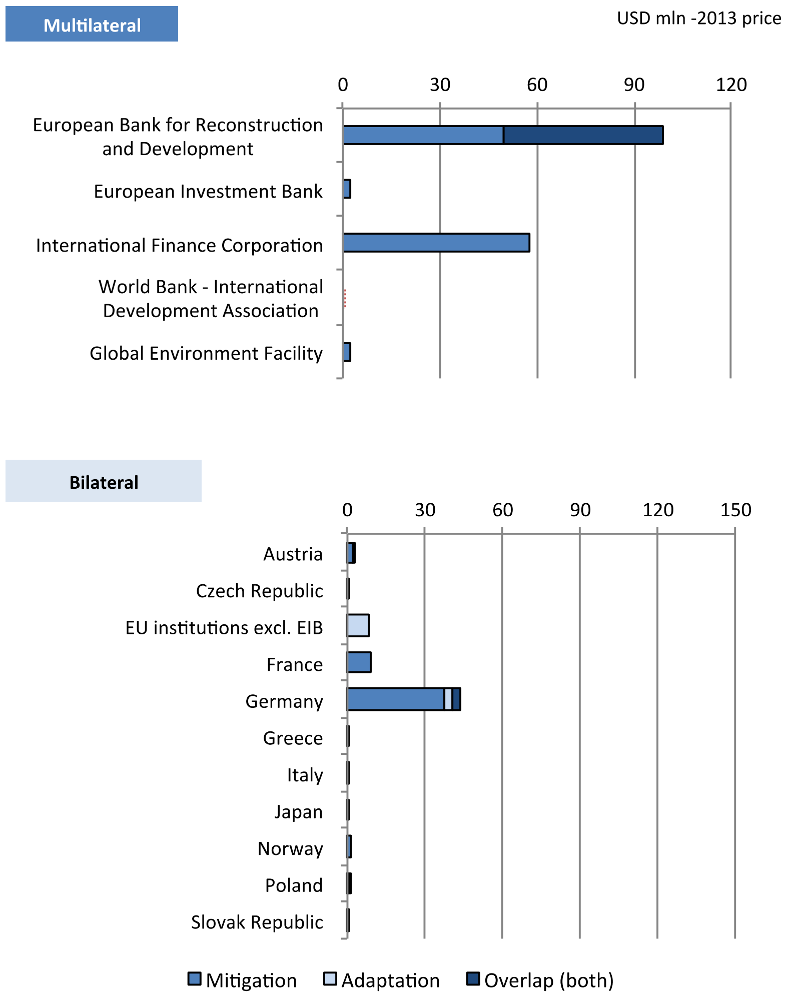
Figure 4. Channels and financial instruments used to deliver climate-related development finance (USD million per year: 2013-14 average)



Source: OECD (2016) Climate-related development finance, 2016 February version.

As mentioned above, the EBRD committed the largest amount of climate-related development finance to the country in the period 2013-14 (Figure 5). A significant portion of the financing targets multi-focal projects (i.e. both mitigation and adaptation), which however is mostly attributed to the hydropower project in Shuakhevi. The EBRD supports projects in various sectors including energy, banking, housing, transport sectors. The IFC, the second largest provider during the period, largely supports hydropower plants, using both non-concessional loan and equity financing. Germany also provides a large amount of finance targeting a diverse set of activities. These include efficient energy generation and transmission, waste management, rural livelihood development and protection of biodiversity. France supports a mitigation project aimed at rehabilitating the aged cable car system in the city of Chiatura.

Figure 5. Major providers of climate-related development finance
(USD million per year: 2013-14 average)



Source: OECD (2016) Climate-related development finance, 2016 February version.

Table 2. Sectoral coverage of mitigation and adaptation projects in 2013 and 2014 by provider
("X" represents that a relevant project (or projects) exists in the sector)

	Agriculture, Forestry and Fishing		Banking, financial and business services		Energy generation and supply		General environmental protection		Government and Civil Society		Health		Humanitarian aid and Developmental Food Aid	
	M	A	M	A	M	A	M	A	M	A	M	A	M	A
Bilateral														
Austria	X				X		X							
Czech Republic					X			X						X
EU institutions excl. EIB		X												
France														
Germany		X		X	X		X	X						
Greece											X	X		
Italy														X
Japan								X						
Norway					X									
Poland	X									X			X	X
Romania														X
Slovak Republic	X	X												
Sweden							X	X						
Switzerland		X												X
United States	X		X	X	X		X	X		X				
Multilateral														
EBRD			X	X	X	X								
European Investment Bank					X	X								
Global Environment Facility	X						X							
International Finance Corporation					X	X								
World Bank - IDA	X	X												

	Industry, Mining, Construction, Trade Policy and Tourism		Multi sector		Other Social Infra-structure		Transport and storage		Unallocated / Unspecified		Water Supply and Sanitation	
	M	A	M	A	M	A	M	A	M	A	M	A
Bilateral												
Austria												
Czech Republic												
EU institutions excl. EIB												
France							X					
Germany											X	
Greece												
Italy												
Japan												
Norway												
Poland												
Romania												
Slovak Republic											X	
Sweden												
Switzerland												
United States	X		X	X								
Multilateral												
EBRD	X	X			X	X	X	X	X	X		
European Investment Bank												
Global Environment Facility			X									
International Finance Corporation	X	X										
World Bank - IDA												

Note: Names of the sectors correspond to those used in the DAC CRS database (OECD, 2016)
Source: Based on OECD (2016), Climate-related development finance, February 2016 version, http://www.oecd.org/dac/stats/climate-change.htm

Selected examples of projects supported by climate-related development finance

This section covers climate mitigation and adaptation activities committed during the period between 2011 and 2015 based on information included in publicly available documents on individual projects/programmes. Each example shows how the activity is financed and what actors are involved in it, both inside and outside of the country. Whilst the previous section uses the OECD DAC statistical data for the period 2013-14, this section rather uses qualitative data with some indicative numbers on each project to illustrate how the country and its development co-operation partners as well as other domestic and international stakeholders are working together to finance climate actions.

Based on the country's intended nationally determined contribution (INDC), the third National Communication, the Technology Needs Assessment report and the draft for the National Energy Efficiency Action Plan (NEEAP), this section outlines selected examples of internationally supported projects on climate change by bilateral and multilateral sources in the following sectors. Some of these projects are also co-financed by domestic sources;

- Energy supply, and energy consumption in cities and buildings;
- Transport;
- Forest ecosystem management;
- Disaster risk management;
- Agriculture;
- Tourism.

Energy supply, and energy consumption in cities and buildings

Georgia's INDC identifies climate actions in the energy sector as priorities for the country including for the pre-2020 mitigation period. The Technology Needs Assessment (TNA) for Georgia conducted under a scheme of the UNFCCC also identifies energy efficiency and renewable energy technologies as priorities for the mitigation efforts of the country (GoG, 2012).

Among the renewable energy sources in Georgia, hydropower represented 74.5% of the total electricity supply and 16.8% of the country's Total Primary Energy Supply (TPES) in 2012. The supply of energy from hydropower has increased by 6.7% since 2002 (IEA, 2015). There have been a number of projects in the country on development and rehabilitation of hydropower plants, including the Shuakhevi plant and plans to install additional 185 MW hydropower capacities. The project has been registered as the project under the Clean Development Mechanism (CDM) of the UNFCCC, aiming to reduce GHG emissions, contribute to private sector investments, and attract investors and support South-South investment, amongst others (IFC, 2014). The EBRD also provides loans to the development of Dariali Hydro Power Plant supplemented by a grant from the Netherlands. The investment is also the first energy-sector project in Georgia with a carbon-neutral construction by including reforestation after the completion of the construction works.

Small-scale hydropower plants also have a significant role to play in contributing to GHG emission reductions of the country. The "Programme on Promotion of Renewable Energies II", supported by Germany and Austria, envisages the promotion of renewable energy sources, particularly small-scale hydro power plants. This programme also aims to enhance the access of private entrepreneurs and enterprises for loan capital. The Bank of Georgia is meant to manage credit lines provided through Germany's KfW and Austria's Development Bank (KfW and OeEB, 2012).

Scaling-up of renewable energies can also benefit from further improvement of the electricity transmission capacities, which is also important for the stability of electricity supply within the country. With improved networks, electricity from renewable sources could also be transported to neighbouring countries such as Turkey, and used to mitigate GHG emissions while generating revenues for Georgia. Germany and the EBRD have supported several projects targeted at the improvement of transmission networks. For instance, the project "Open Program Transmission Network II" aims at increasing power transmission capacities, improving security and quality of supply and network access of renewable energies, especially electricity by hydropower. This project is also complemented by the "Jvari-Khorga Interconnection" project that is financed by the EBRD and aimed to strengthen the reliability and stability of the Georgian transmission network. Germany and the EBRD, together with the European Investment Bank (EIB) and the EU also finance the Black Sea Transmission Network project.

Apart from hydropower, the development of wind, biomass, solar and geothermal technologies has a great potential to reduce Georgia's GHG emissions and contribute to the diversification of energy sources (GoG, 2012). For instance, an increase in biomass use for heating can contribute to GHG emission mitigation in two ways: by reducing combusting natural gas and by reducing the leaks from the natural gas grids in the country. Biofuels and solid waste are the next largest source of renewable energy to hydropower, with a share of 8.3% in TPES, and are mainly used in households for heating and cooking purposes. However, energy from biofuels and waste has declined by 52.5% since 2002 (IEA, 2015). The "Promotion of Biomass Production and Utilization in Georgia" project, supported by GEF and co-financed by the Georgian government, was designed to promote the production and use of upgraded biomass fuels to meet the needs of the municipal services sector for heat in a sustainable and efficient way, thereby reducing the country's dependence on fossil fuels and avoiding GHG emissions (UNDP, 2011).

The share of fugitive GHG emissions (i.e. natural gas) was also quite high (28% in 2011) compared with many other EECCA countries, and thus the government considers the reduction of the leaks of natural gas also to be a priority. The GEF-supported project on Promotion of Biomass Production and Utilization mentioned above also contributes to reducing the fugitive gas from the natural gas grids in the major cities. Another example, albeit not public finance, is a CDM project that aims to reduce leaks at above-ground infrastructure in the distribution system (e.g. valves and cranes) of Socar Georgia Gas (Socar Georgia Gas & Arcadia Energy, 2012).

Improving energy efficiency on the demand side (e.g. residential and commercial building and manufacturing sectors) can contribute significantly to GHG emission reductions. For instance, assuming 20% of GHG emission reduction below business as usual (BAU) towards 2030, 37% of total savings could be achieved in the residential sector, 23% in the industry sector, 15% in the transport sector, and 11% in the commercial sector (GoG, 2015b). Infrastructure in Georgia is often characterised as aged, over-designed in the Soviet-era and thus expensive to maintain, and inefficient in its energy use.

"The Second Regional and Municipal Infrastructure Development Project", supported by the World Bank and Switzerland, aims to improve the efficiency and reliability of targeted municipal services and infrastructure (e.g. water supply and sanitation as well as city transports). Examples of measures to be taken include: use of gravity-fed water supply or replacement of old pumps with energy efficient ones, replacement of street lighting with more energy efficient bulbs, and improvement of heating efficiency in municipal buildings, especially kindergartens (WB, 2014). The "EC-LEDS Clean Energy Program in Georgia", supported by the United States, is a technical assistance project aiming to: help Georgian municipalities institutionalise and implement climate change mitigation measures, promote and facilitate private sector investments in energy efficiency and green buildings; and build the capacity of the government to develop and implement a national Low Emissions Development Strategy (LEDS). The programme also supports the Georgian government in promoting investments in new hydropower plants and establishing an energy trading mechanism (USAID, n.d.).

The EIB and the EBRD, together with Germany, the Netherlands, Austria and the IFC, invested in the "Green for Growth Fund (GGF)" that aims to enhance the use of energy efficiency measures and renewable energy sources in Georgia as well as Southeast Europe and other EU's Eastern Neighbourhood countries[3]. The GGF provides refinancing to financial institutions for on-lending to enterprises and private households seeking to finance energy efficiency projects. The GGF also invests directly in small to medium-scale renewable energy projects. In Georgia, the GGF provides finance to the Bank of Georgia to promote the construction of energy efficient housing initially in the Tbilisi area (GGF, 2014).

Table 3. Examples of projects supported by international climate-related development finance (Committed in 2011-2015)

Project type	Project	Finance provider	Financial instrument and amount	Co-financing by domestic actor	Key domestic institution
Hydro-power	The Shuakhevi plant (2013-14)	IFC, EBRD, ADB, Canada	Non-concessional Loan (IFC: USD 70 mln, EBRD: USD 90 mln, ADB: USD 75 mln) Loan* (Canada: USD 15 mln) Equity (IFC: USD 34 mln)	N.A.**	Ministry of Energy
	Program Promotion of Renewable Energies II	Germany, Austria	Concessional loan (Germany: USD 33.9 mln) Grant (USD 0.6 mln)	N.A.**	JSC Bank of Georgia
	Dariali Hydro Power Plant	EBRD, Netherlands	Non-concessional loan (EBRD: USD 40mln) Loan* (NL: 30 mln, GGF: 10 mln***)	JSC Dariali Energy (but amount not found)	JSC Dariali Energy, The Georgian Energy Development Fund
Biomass energy and fugitive gas reduction	Biomass Production and Utilization in Georgia (2011)	GEF, UNDP	Grant (GEF: USD 0.93 mln, UNDP: USD 0.16 mln)	GoG (USD 3 mln), Private sector (USD 1 mln)	Ministry of Environment Protection and Natural Resources
Wind power	Gori Wind (2015)	EBRD	Non-concessional loan (USD 24 mln)	N.A.**	Georgian Energy Development Fund
Energy efficiency in trans-mission	Open Program - Extension Transmission Network	Germany (through KfW)	Concessional Loan (USD 46.5 mln)	N.A.**	Georgian State Electrosystem JSC
Energy efficiency at demand side	Second Regional and Municipal Infrastructure Development Project (2014)	WB (IBRD), Switzerland	Concessional loan (IBRD: USD 30 mln) Grant (Switzerland: USD 5 mln)	GoG (USD 11 mln), Local financial institutions (USD 12.9 mln)	Municipal Development Fund of Georgia, Local self-governments

[3] Southeast Europe includes Albania, Bosnia and Herzegovina, Croatia, Former Yugoslav Republic of Macedonia, Kosovo*, Montenegro, Serbia and Turkey, as well as in the nearby EU's Eastern Neighbourhood region comprised of Armenia, Azerbaijan, Georgia, Moldova and Ukraine. (*This document is without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area).

	EC-LEDS (Enhancing Capacity for Low Emission Development Strategies) (2013)	United States (through USAID)	Grant (USD 2.7 mln)	N.A.	Ministry of Environment Protection and Natural Resources, Ministry of Energy, Ministry of Economy and Sustainable development, NGOs in Georgia, municipalities
	Financing for the construction of energy efficient housing by GGF	Germany (KfW), EIB, and EBRD through the Green for Growth Fund (GGF)	Loan* (EUR 15 mln)	N.A.	JSC Bank of Georgia

* Information on concessionality is not found.
** N.A. Information on co-financing from domestic actors is not found.
***GGF: Green for Growth Fund, Southeast Europe.

Transport

As stressed in the Second National Communication and reiterated in the Third Communication of Georgia to the UNFCCC, improving efficiency in the transport sector and developing sustainable transport systems are priorities for the Georgia's cost-effective GHG emission reductions (GoG, 2015b; GEF&UNDP, 2014). The growth in transport demand is already reflected in the increase in consumption of imported oil products (GoG, 2015b). The Technology Needs Assessment for mitigation in Georgia also identifies multiple priority areas for the transport sector, such as the promotion of public transport systems in cities, introduction of electric, hybrids and plug-in hybrids, increasing the use of biofuels (GoG, 2012),.

There are several barriers identified to developing low-carbon and climate-resilient transport systems. For instance, Georgia's transport sector often lacks: local governments' capacities; institutional exposure to the best international practices of sustainable transport systems; access to finance; and public awareness to support and increase demand for such transport systems. "Green Cities: Integrated Sustainable Transport in the City of Batumi and the Achara Region", supported by the GEF and the UNDP, aims to help the local governments (e.g. City of Batumi) develop and adopt urban transport plans and also increase investment in less-GHG intensive transport and urban systems (GEF&UNDP, 2014).

The "Sustainable Urban Transport Investment Program", supported by the ADB as an 8-year multi-tranche financing facility aims to support the rehabilitation of the public transport network and develop efficient transport systems in key urban areas. Sub-projects under the programme include mass transit, essential road links, and coastal protection in Georgia. The project was not recorded in the OECD DAC CRS database in 2013 or 2014, while it seems to at least partly contribute to the GHG emission reduction against the business-as-usual baseline. France also supports rehabilitation of ropeways in the municipality of Chiatura (Imereti region), which includes a climate mitigation objective (OECD, 2016).

Table 4. Examples of projects supported by international climate-related development finance (Committed in 2011-2015)

Project type	Project	Finance provider	Financial instrument and amount	Co-financing by domestic actor	Key domestic institution
Efficient transport system	Green Cities: Integrated Sustainable Transport for the City of Batumi and the Achara Region	GEF, UNDP	Grant (USD 0.85 mln) In-kind (UDNP: USD 0.2 mln)	City of Batumi (USD 10.3mln) GoG (USD 0.1 mln)	Ministry of Environment and Natural Resources Protection, City of Batumi
	Sustainable Urban Transport Investment Program (Tranches 2 in 2012, 3 in 2013 and 4 in 2014)	ADB	Loan* (USD 141 mln)	GoG (USD 49.2 mln)	The Municipal Development Fund
	Rehabilitation of ropeways in Chiatura	France (through AFD)	Concessional loan (USD17.4mln)	N.A**.	Ministry of Regional Development and Infrastructure

* Information on concessionality is not found.
**Information on co-financing from domestic actors is not found.

Forest ecosystem management

Georgia's INDC spends one section on forestry, while the country has not decided whether it includes land use, land use change and forestry (LULUCF) in its GHG reduction targets to 2030 (GoG, 2015a). Nevertheless, the INDC states that measures on afforestation and reforestation, as well as sustainable forest management, are planned to be implemented, although the extent to which such measures will be taken will depend on the level of international support. Table 5 provides examples of some measures outlined in Georgia's INDC to address the issues on LULUCF. These measures include both commitments conditional on international financial and technical support, and unconditional commitments.

Some of the forest covers are also vulnerable to climate change, and thus good management of forests can improve their adaptive capacity to on-going and future changes in climate. It has been already observed that the increase in temperature and precipitation in the last few decades has led to the annual increase in the prevalence of pests and diseases in forests in the country (GoG, 2015b).

Table 5. Measures on LULUCF included in Georgia's INDC

Conditionality	Measures
Unconditional commitments	• Support CO_2 reduction in the Borjomi-Bakuriani Forest district by at least 70% between 2020 and 2030 by strengthening law enforcement and introducing Sustainable Forest Management practices • Afforest 1 500 ha of degraded non-treed land by 2030 • Assist the natural regeneration of forests through fencing of 7 500 ha by 2030 in order to restore natural forest cover
Conditional commitments on necessary financial and technical support from international sources	• Afforest/reforest up to a total of 35 000 ha as well as support relevant activities to assist natural regeneration in identified areas needing afforestation / reforestation until 2030 • Support the sustainable management of forests with estimating measures leading to an overall carbon sequestration up to 6 million tons of CO_2 on these lands over the period 2020-2030 • Expand the protected area from 0.52 million ha to 1.3 million ha (about 20% of Georgia's territory) comprising at least 1 million ha of forests

Source: GoG (2015a) Georgia's intended nationally determined contribution submission to the UNFCCC.

The "Sustainable Management of Biodiversity, South Caucasus" project, supported by Austria, was designed to improve forest management in a sustainable manner. The actors involved include the institutions that engage in forest management, forest policy development, and monitoring and supervision. The final beneficiary is the rural population in the country. The project also aims to help the country establish a National Forest Monitoring Systems (NFMS) based on remote sensing data that provides reliable information on the status and development of the forests (ADC, 2010).

Measures to stop unsustainable forest management require sufficiently accurate information on forests and land use. However, Georgia (like many other developing countries) lacks access to credible, independent and regularly updated information on the status of forests (GEF, 2013). The project "Global Forest Watch 2.0" supported by GEF, the World Resources Institute, a range of bilateral donors and private sector actors, helps to ensure the information systems effectively engage with the wider public and achieve optimum transparency and credibility. The project aims also to introduce forest and land management tools that can support the development and implementation of collaborative cross-sectoral integrated land use management plans (GEF, 2013).

Table 6. Examples of projects supported by international climate-related development finance (Committed in 2011-2015)

Project type	Project	Finance provider	Financial instrument and amount	Co-financing by domestic actor	Key domestic institution
Forest-related policy development	ADA-BMZ/GIZ Forest Sector Reform Programme in Georgia (2014)	Austria (through GIZ)	Grant (USD 2.4 mln)	N.A.*	Ministry of Environment and Natural Resource Protection
Development of evidence base	Global Forest Watch	GEF, bilateral donors**	Grant** (USD1.8mln)*	GoG (USD2mln)	Ministry of Environment and Natural Resource Protection

* N.A.: Information on co-financing from domestic sources is not found.

**There are multiple bilateral public providers of finance and private sector investors involved in the programme. For further details, see (GEF, 2013).

Disaster risk management

Climate change is likely to amplify the risks of major natural disasters, and the Third National Communication mentions that the establishment of early warning systems for climate-related extreme events is a priority (GoG, 2015b). Hazard warning systems are already put in place in the country, although they still need further improvements in terms of reliability, credibility and consistency (UNDP, 2015).

The Adaptation Fund supported the project on "Developing Climate Resilient Flood and Flash Flood Management Practices to Protect Vulnerable Communities of Georgia". The project was designed to help the governments and the population of the Rioni Basin to enhance their adaptive capacity and resilience of the economy. The project focuses on floodplain development policies to promote long term resilience to flood/flash flood risks, climate resilient practices of flood management in highly exposed communities, and the establishment of early warning systems (AF, 2011). Switzerland, Poland and the Czech Republic, among others, also support disaster risk reduction in Georgia. For instance, Poland-supported "Anti-flood early warning and prevention systems" project aims to support the communities around the Kabali and Duruji rivers with hydro gauge installation, student training, hydro meteorological modelling and early warning systems (Civitas Georgica, 2014). Switzerland supports the government's disaster management efforts through the introduction of methodologies of risk mapping and cost benefit analysis. It also provides vocational training to municipal fire fighters and rescuers (EDA, 2016).

Table 7. Examples of projects supported by international climate-related development finance (Committed in 2011-2015)

Project type	Project	Finance provider	Financial instrument and amount	Co-financing by domestic actor	Key domestic institution
Early warning system	Developing Climate Resilient Flood and Flash Flood Management Practices to Protect Vulnerable Communities of Georgia	Adaptation Fund, UNDP	Grant (AF: USD 4.9 mln, UNDP: USD 0.16 mln)	N.A.*	Ministry of Infrastructure and Regional, Development, Emergency Management Department, Pilot municipalities
	Anti-flood early warning and prevention systems in Georgia: Special focus on the Kabali and Duruji rivers	Poland	Grant (USD 0.2 mln)	N.A.*	National Environmental Agency, NGO “Civitas Georgica”
Risk mapping and training	Strengthening the prevention and preparedness systems	Switzerland (through EDA)	Grant (USD 0.67 mln)	N.A.*	Ministry of Environment and Natural Resource Protection, Ministry of Internal Affairs, municipalities on local level

* N.A.: Information on co-financing from domestic sources is not found.

Agriculture

The Third National Communication of Georgia to the UNFCCC examines the vulnerability of the agriculture sector to climate change and states that the sector is a priority in the country’s adaptation efforts. In addition, studies on climate change predictions for Georgia estimate a possible increase by 3.5°C in mean annual temperature by the end of this century, accompanied with a decrease in precipitation by about 6% in the western regions (IFAD and GEF, 2012). This can have a significant implication for water use of the agriculture sector where estimated water efficiency of an operational irrigation system is less than 40% (ibid.).

Multiple projects have been implemented in this sector with support of the EU under the “European Neighbourhood Programme for Agriculture and Rural Development (ENPARD)” that builds on the EU’s best practices in developing agricultural and rural areas. The ENPARD’s overarching objectives include improving employment and living conditions in rural areas through the diversification of the rural economy. In the context of climate change adaptation, the programme aims to support the adoption of more competitive and climate resilient agricultural practices, promotion of viable off-farm economic activities for income source diversification, and sustainable management and protection of environmental resources in rural areas (Ortega, 2014). The pilot projects are conducted in, for instance, the municipalities of Lagodekhi, Kazbegi and Borjomi.

As mentioned above, water resource management relating to irrigation systems can also be greatly affected by climate change. The IFAD and the GEF support “Enhancing Resilience of Agricultural Sector in Georgia” project that aims to improve water availability, farmland productivity and smallholders’ income through investments in climate-resilient farming systems (IFAD and GEF, 2012). The “Irrigation and Land Market Development Project" supported by the World Bank aims to: increase the absolute volume of irrigation water supplied; expand areas provided with improved irrigation and drainage services; and recommend policies and procedures for a national programme of land registration. The project also helps to formulate a National Irrigation and Drainage Strategy that would define approaches that the country could take in order to enhance resilience of the agriculture sector to climate change (WB, 2014b).

An effective agriculture insurance scheme (e.g. crop insurance) could play an important role to help farmers better financially prepare for climate risks by easing their economic losses when natural disasters occur. About 90% of the rural population in Georgia is involved in small scale subsistence or semi-subsistence agriculture (EDA, 2011). They are particularly vulnerable once a natural disaster occurs and damages their assets. Inadequate methodologies of value chain financing for agricultural activities and existing land fragmentation make it more difficult for small farmers to access affordable crop insurance. Under the "Market Alliances in the Lesser Caucasus Region of Georgia" Programme, Switzerland helps the small-sized actors in the agriculture sector with better access to finance and agriculture insurance (EDA, 2011).

Table 8. Examples of projects supported by international climate-related development finance (Committed in 2011-2015)

Project type	Project	Finance provider	Financial instrument and amount	Co-financing by domestic actor	Key domestic institution
Resilient rural development	European Neighbourhood Programme for Agriculture and Rural Development (ENPARD) (2012 and 2014)	EU	Grants (USD 53.8 mln in 2012 and USD 15.8 mln in 2014)	N.A.*	Ministry of Agriculture, Georgian Farmers Association, Municipalities of Lagodekhi, Kazbegi and Borjomi
Agriculture insurance and facilitating market access	Market Alliances in the Lesser Caucasus Region of Georgia (2014)	Switzerland (through Swiss Development Cooperation)	Grant (USD 9 mln)	N.A.*	Ministry of Agriculture
Irrigation and drainage management	Enhancing Resilience of the Agricultural Sector in Georgia (2015)	IFAD, GEF	Loan** (IFAD: USD 13.5 mln) Grant (GEF: USD 5.3 mln, IFAD: USD 0.2 mln)	GoG (USD 2.1 mln)	Ministry of Agriculture
	Irrigation and Land Market Development Project (2014)	WB (IDA)	Concessional loan (USD 0.9 mln)	N.A.*	Ministry of Agriculture, Ministry of Justice

* Information on co-financing from domestic sources is not found.
** Information on concessionality is not found.

Tourism (the Black Sea coastal zone protection)

The Third National Communication shows the result of climate risk assessment for the period until 2100, which predicts that "further deterioration of favourable conditions for tourism in the coastal area (around the Black Sea) was projected for July-August, which will lead to the need to implement relevant adaptation measures" (GoG, 2015b). This is also highlighted in the INDC and in the absence of adaptation measures, the estimated economic losses in the tourism sector are predicted to be about USD 2 billion by 2030. The INDC stresses the importance of integrated coastal planning and management instruments, rather than investments in coastal erosion abatement only (GoG, 2015a). The Technology Needs

Assessment also selected land degradation caused by water erosion and natural disasters in the ecosystems in the Black Sea coastal zone as the highest priorities, particularly in the tourism development areas and the free economic zones (GoG, 2012)

The EBRD has conducted a pilot project in Georgia aimed at integrating a climate change assessment into Environmental and Social Impact Assessments. This project and the climate change assessment aim to provide recommendations on adaptation measures that need to be taken into account in a further investment related to the expansion of a major port on the Black Sea coast. These recommendations are expected to inform project development for on the port that is sensitive to climate change impacts such as potential sea level rises and changes in sedimentation patterns due to climate affected glacial rivers (EBRD, 2009). However, no particular project with support of international climate finance committed in 2011-2015, focusing specifically on coastal zone climate change adaptation, has been identified.

In-country enabling environments for climate actions

Legal and policy frameworks

Georgia has developed several policy documents to pursue low-carbon, climate-resilient development. For instance, the Law on Electricity and Natural Gas includes objectives related to improving the electricity and natural gas markets and tariff systems and making them efficient and competitive, as well promoting the use of indigenous hydro energy, other renewable, alternative and natural gas resources. The Strategy for Regional Development for 2010-2020 (Resolution of the Government No.172) considers climate change adaptation and sustainable development important national objectives.

The Low Emission Development Strategy (LEDS) is now being developed and is planned to be finalised by the autumn of 2016, with support of the United States. The LEDS aims, amongst others, to improve national energy security by facilitating investments in new hydropower plants and establishing an energy trading mechanism. The National Energy Efficiency Action Plan (NEEAP), with support by the EBRD, is also in a process of development. This NEEAP is to identify energy efficiency improvement measures and expected energy savings in all sectors (e.g. buildings, transport, power generation, industry and services). Three NAMAs are also being developed and meant to be linked with the other overarching policy documents such as the NEEAP.

LEDS and NEEAP are also highlighted in the INDC as means to identifying Georgia’s pre-2020 actions. Moreover, the country also plans to develop “Climate 2021-2030” by 2018, which will define the legal instruments, activities, methods and other relevant issues, and with the aim of co-ordinating climate related multi-sectoral activities in the country and provide pathway for reaching the country’s climate targets. (GoG, 2015a; Mdivani, 2016).

Table 9. Summary of policy instruments (renewable energy and energy efficiency)

Regulatory policies for renewable energies		
	Renewable energy targets	
	Biofuels obligation / mandate	
	Electric utility quotas obligation / Renewable Portfolio Standard (RPS)	
	Feed-in tariff / premium payments	X
	Heat obligation / mandate	
	Net metering	
	Tendering (i.e. Public bidding) for renewable energy	
	Tradable renewable energies certificates	
Fiscal incentives for renewable energies and public financing		
	Capital subsidy / rebate	
	Energy production payment	
	Investment or production tax credits	
	Public investment, loans or grants	X
	Reduction in sales, energy, CO_2, VAT or other taxes	
Energy efficiency policies		
	Energy efficiency target	
	National energy efficiency awareness campaigns	X
	National energy efficiency regulations, standards or laws	
	Governmental institution(s) to formulate and implement energy efficiency strategies and policies	X
	Energy efficiency labelling policies	
Adaptation		
	National-level policy document that facilitates adaptation	X

Source: Based on the data from UNECE and REN21 (2015) "The UNECE Renewable Energy Status Report"; and GoG (2015a) Intended Nationally Determined Contribution (on adaptation).

Domestic climate finance mechanisms and frameworks (selected examples)

As discussed in the previous section, Georgia has co-financed a range of climate-related projects using its own domestic resources. Georgia has established mechanisms and policy frameworks to promote the mobilisation of domestic public and private climate finance sources. These include, among others, national funds that can be used to allocate resources to climate actions, reforms on tariffs for electricity, and public budget allocation to low-carbon technologies and capital investments. Below we show some examples of such mechanisms and frameworks.

Tariffs

Georgia has established an incentive scheme for small hydropower plants. The average generation tariff in Georgia is approximately USD 0.028/kWh with a variation of USD 0.007/kWh for older and USD 0.068/kWh for newer plants. However, small hydropower plants, unlike large-scale ones, are not required to sell their electricity to the national grid, but directly to consumers at bilaterally negotiated tariffs (UNDP, 2010).

Georgian Energy Development Fund

The Georgian Energy Development Fund (GEDF) is a joint stock company established in 2010 by the Government of Georgia in order to facilitate investment and development of the country's renewable energy sector (GEFD, 2014). GEDF identifies new renewable energy projects and carries out pre-feasibility assessment of projects. Such projects are offered to potential investors (with or without GEDF co-investment). GEDF has invested in 10 projects since 2011: the total amount of investments is over USD 20 million (GEDF, n.d.). More than 20 projects are currently at various stages of development.

Capital investment by the government

Georgia has allocated, and been planning to allocate, state budget and municipal budget resources to a range of climate-related projects in the country. The government has also estimated the investment needs, such as for the power sector and demand-side devices, in its Third National Communication to the UNFCCC. Such costs include those of fuel extraction and import, operating and maintenance costs of the energy sector infrastructure, and investments in new power plants, and the purchase of new demand-side devices (GoG, 2015b). Further, the draft National Environmental Action Plan of Georgia for 2011-2015 (NEAP) outlined several measures to be at least partly financed by the domestic public budget (GoG, 2010). The following measures were included in the draft NEAP:

- Investment to strengthen the coastal zone infrastructure at the delta of the Rioni River and the Batumi-Adlia area;
- Research on potential risks of climate change impacts in the Eastern Georgian semi-arid regions;
- Research on the current state of existing irrigation systems in the Eastern Georgia's semi-arid regions;
- Rehabilitation of irrigation systems in the Dedoplistskaro region;
- Preparation for the annual National GHG Inventory;
- Assessment of GHG emissions from the city of Tbilisi for 2020 (using municipal budget resources);
- Implementation of the strategy for emission reduction from the established baseline under the "Covenant of Mayors" Initiative (using municipal budget resources);
- Awareness-raising campaigns on potential climate change risks;
- Implementation of a study on possible impacts of climate change on the mountainous regions (Khevi, Adjara, Upper Svaneti);
- Preparation of measures to adapt to adverse impacts of climate change on the healthcare and tourism sectors;
- Planting mudflow-breaking forests (hazelnuts).

Further, the National Energy Efficiency Plan is being developed to identify significant energy efficiency improvement measures and expected energy savings in all sectors, and meant to be published in 2016. The third National Communication outlines 20 energy efficiency measures under the Plan with estimated costs to be financed by domestic and international financial sources. The total estimated costs is approximately EUR 1.6 billion (GoG, 2016).

Annex: Key institutions and legal and policy frameworks

The tables below show key institutions and legal and policy frameworks in the country that are, or will be, related to accessing and using climate-related development finance. The institutions include: those engaged in development planning; those in charge of environmental policies and regulations; those which manage or oversee energy industry; those which are private or state-owned entities engaged in work with international climate finance sources; those whose work is related to adaptation (e.g. water, disaster risk management, etc.).

Major domestic institutions involved in climate-related projects in the country

Entity	Description
Ministry of Environment and Natural Resources Protection	Responsible for elaborating and implementing policies of the Georgian Government in the areas of sustainable development, climate change, environmental protection and sustainable use of natural resources
Ministry of Energy	In charge of regulating the activities in the energy sector
Ministry of Economy and Sustainable Development	In charge of development and implementation of policies related to the provision of incentives for economic growth, sustainable development and green economic growth of the country
Ministry of Agriculture	In charge of regulation of economic activity in the agricultural sector of the country with a purpose of increasing the sector's production capacity
Ministry of Labour, Health and Social Affairs	In charge of regulating the health care system, labour issues and social security system
Ministry of Regional Development and Infrastructure	In charge of regional and infrastructure development
Ministry of Internal Affairs	In charge of law enforcement, divided in several sub-branches that include the Emergency Management Agency
National Forestry Agency	Responsible for planning and implementing the state policy in the field of forest fund protection, restoration and renewal of forests and usage of forest resources amongst others
National Environmental Agency	In charge of, for instance, preparing and spreading informational documents, forecasts, warnings regarding the existing and expected hydro-meteorological and geodynamic processes
Governmental Coordination Committee on Low Emissions Development Strategy	In charge of supporting decision-making on Low Emissions Development Strategy, consisting of high-level representatives (Deputy Ministers) of climate-relevant national ministries and agencies
Coordination Council for Implementation of CDM	In charge of co-ordinating activities related to the Clean Development Mechanism under the Kyoto Protocol
Municipal governments	Those that are members of the Covenant of Mayors
Municipal Development Fund	In charge of supporting institutional and financial capacity strengthening of local government units and investing in local infrastructure and services
Georgian Energy Development Fund	Facilitating investments and development of the country's renewable energy sector

Sources: Based on GoG (2016), First Biennial Update Report on Climate Change; GoG (2015b), The Third National Communication to UNFCCC.

Major legal and policy documents relevant to climate action (Examples)

Name	Description
Law on Electricity and Natural Gas	The objectives of this Law are to: • Regulate the existing non-competitive market, establish electricity and natural gas markets and tariff systems in an efficient and competitive way; • Provide the legal basis for reliable electricity and natural gas for all categories of consumers; • Encourage domestic and foreign investment; and • Encourage the use of indigenous hydro energy, other renewable, alternative and

	natural gas resources.
Resolution of the Parliament of Georgia on "Main Directions of State Policy in the Power Sector of Georgia" №3758 -II	This is a supporting document aimed at the promotion of climate change mitigation and the enhancement of renewable energy sources, and in particular of the hydropower energy sector as well as wind energy development. This document establishes specific targets and describes the power sector potential.
Strategy for Regional Development for 2010-2020 Resolution of the Government №172, 25.06.2010	The Strategy is a cross-sectoral national policy document, which considers climate change adaptation and sustainable development issues major policy issues. The Strategy also promotes the use of renewable energy sources, but does not mention their link to climate change mitigation.
Strategy for Agriculture Development for 2015-2020	The new strategy document contains main strategic directions of development of agriculture in Georgia. The main directions include 'climate, environment and biodiversity protection' that implies, among others, support to the implementation of climate-smart agricultural practice in Georgia.
The Second National Environmental Action Programme (NEAP-2)	NEAP-2 covers both environmental (land, bio-resources, water, air, underground resources, hazardous waste and substances) and cross-cutting issues (environmental impact assessment and permitting, enforcement, environmental education and public awareness, monitoring, the scientific basis for decision-making and the need for geographic information, disasters prevention measures, etc.).
Low Emission Development Strategy (LEDS -draft)	Working on the Low Emission Development Strategy (LEDS) began late in 2012 and is expected to be finalised by the autumn of 2016. The project is based on the Memorandum of Understanding between the Governments of USA and Georgia on cooperation in the LEDS area under the USA EC LEDS programme (Enhancing Capacity in LED Strategy).
National Energy Efficiency Action Plan (NEEAP -draft)	As of July 2016, the NEEAP is being developed to identify energy efficiency improvement measures and expected energy savings in all sectors (e.g. buildings, transport, power generation, industry and services).

Sources: Based on GoG (2016), First Biennial Update Report on Climate Change; GoG (2015) Intended Nationally Determined Contribution; IEA (2015), Energy Policies Beyond IEA Countries: Caspian and Black Sea Regions 2015, IEA Publications, Paris; Clima East (2016) Georgia: Country Profile (at 1 January 2016), http://www.climaeast.eu/ .

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